

Shima Nabiee

Contact details

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Education

- **Ph.D./M.Sc. - Electrical and Computer Engineering** 2015 - 2023
University of California, Irvine
- **B.Sc. - Electrical Engineering** 2015
Sharif University of Technology, Iran

Skills

- **Generative AI:**
 - LLMs, VLMs, SLMs (GPT-4, Mistral, LLaMA, CLIP, BLIP-2, DALL-E)
 - Prompt engineering, in-context learning
 - Fine-tuning (LoRA, QLoRA, PEFT, adapters)
 - RAG pipelines
 - Diffusion models
 - Model optimization (quantization, pruning, efficient inference)
 - Agent frameworks (LangChain, OpenAI Function Calling, ReAct)
- **Programming & Tools:**
 - Python, Git, experience with C++ and JavaScript
 - PyTorch, HuggingFace Transformers, TensorFlow, Keras, Scikit-learn, OpenAI API
 - Weights & Biases, MLflow
- **Data & Infrastructure:**
 - Multimodal and time-series data pre-processing
 - MySQL, PostgreSQL
 - Cloud platforms: AWS (including Bedrock), GCP, Lambda Labs, Azure

SUMMARY

Generative AI expert with a PhD from UC Irvine (2023) and 10 years' hands-on experience in **computer vision**, **NLP**, and **Python**. Specialized in **LLM**, **VLM**, and **multimodal pipelines** (fine-tuning, model optimization, **RAG**, few-shot learning), **LLM orchestration**, **agent systems**, and **end-to-end AI infra (IaaS)**. Led high-impact **R&D** and consulting, including **foundation model benchmarking** and AI infra design that enabled **multi-million-dollar investments**. Delivered solutions in **custom LLMs**, **facial attribute analysis**, and **enterprise GenAI platforms**. Proficient in **PyTorch**, **TensorFlow**, **Hugging Face**, **OpenAI API**, **SQL**, and ML tooling across **AWS**, **GCP**, **Azure**, **Docker**, and **Lambda Labs**; familiar with modern libraries and workflows for **transformers**, **diffusion models**, **vision-language systems**, and **inference optimization**, etc.

EXPERIENCE

- **AlgoFace – NSF-Funded Startup**
 - *Computer Vision Scientist* Since Sept 2024
 - Researched novel, state-of-the-art approaches and developed a facial attribute analysis system with semantic search, integrating **33 attribute classifiers** and **embedding-based retrieval**.
 - Generated a synthetic training dataset using **DALL-E + GPT-4**, conditioning on descriptive prompts to represent 33 facial attributes.
 - Analyzed attribute correlations and fairness sensitivity to guide **multitask vs. single task** modeling and optimize learning across correlated or sensitive groups.
 - Designed and led a multimodal classification pipeline using **CLIP (ViT)**, **GPT-4**, and **DALL-E**, outperforming baseline models (**VGG**, **ViT**, **FairFace**, **FaceXFormer**).
 - Fine-tuned LLMs using **fair contrastive**, **shape-aware**, and **edge-aware losses**, achieving **85–98.5% accuracy** with **<500 low-res training samples**.
 - Validated against human-labeled data, achieving **5–12% higher accuracy** and enabling **significant cost savings**.

- Pandita AI**

Generative AI Research Scientist

 - Led AI innovation for the communications division of a multinational conglomerate, designing all layers of the **AI Stack** across **AI-focused IaaS** and **PaaS**, as part of a company-wide **AI cloud initiative**.
 - Conducted deep research into AI hardware performance across **A100, H100, B100, V100, MI300, WSE-3**, guiding GPU selection for enterprise workloads.
 - Delivered **strategic recommendations** that led to a **multi-million-dollar investment** in advanced AI infrastructure.
 - Researched a broad range of **LLMs, SLMs**, and **multimodal models** to recommend core models for the initial **AI Stack**, propose future options, and identify partnership opportunities.
 - Analyzed competing AI cloud providers and proposed a feature roadmap—defining launch-critical capabilities and phased additions aligned with user growth.
 - Assessed platforms like **OpenAI, Databricks**, and **Hugging Face** for partnership potential and ecosystem fit.

Palo Alto, CA

Jan. - July. 2024

- Rootick**

Research Engineer

 - Designed and implemented an intuitive, search-optimized website UI for a structured product directory, emphasizing semantic clarity and multi-layered navigation.
 - Developed and maintained a custom **web scraping pipeline** to extract structured and unstructured data across diverse online sources and APIs.
 - Created and curated a large-scale dataset from scraped content, supporting deep **multi-level product categorization** (10+ layers) and taxonomic mapping.
 - Built a custom **LLM-based pipeline** to analyze extracted data, enabling automated **semantic classification** and contextual summarization of product domains.

Sacramento, CA

Aug 2023 - July 2024

PUBLICATIONS

- Nabiee, Shima, Matthew Harding, Jonathan Hersh, and Nader Bagherzadeh. *"Hybrid U-Net: Semantic segmentation of high-resolution satellite images to detect war destruction."* Machine Learning with Applications (2022): 100381.
- Nabiee, Shima, Nader Bagherzadeh. *"Stock Trend Prediction: A Semantic Segmentation Approach."* Submitted to the Multimedia Systems Journal.
- Nabiee, Shima. *From Satellite Images to Financial Time-Series: Using Semantic Segmentation in Real World Applications."* (Ph.D. thesis)
- Nabiee, Shima. *Prediction of Firms' Annual and Quarterly Returns Using NLP Techniques.* (M.Sc. thesis)

HONORS AND AWARDS

- Recipient of the Division of Teaching Excellence and Innovation (DETI) Fellowship, 2021
- Awarded the Henry Samueli School of Engineering Fellowship, 2016